



TIPS & TRICKS



Clearing the terminal screen

Enter 'clear' without quotes in the terminal and hit the Enter button. This causes the screen to be cleared, making it look like a new terminal.

—Abhinav B, ohyesabhi2393@gmail.com



Convert image formats from the command line in Ubuntu

'convert' is a command line tool that works very well in many Linux based OSs. The 'convert' program is a part of the ImageMagick suite of tools and is available for all major Linux based operating systems. If it is not on your computer, you can install it using your package manager.

It can convert between image formats as well as resize an image, blur, crop, dither, draw on, flip, join, and re-sample more from your command line.

The syntax is as follows:

```
convert [input options] input file [output options] output file.
```

For example, we can convert a PNG image to GIF by giving the following command:

```
convert image.png image.gif
```

To convert a JPG image to BMP, you can give the following command:

```
convert image.jpg image.bmp
```

The tool can also be used to resize an image, for which the syntax is shown below:

```
convert [nameofimage.jpg] -resize [dimensions]  
[newnameofimage.jpg]
```

For example, to convert an image to a size of 800 x 600, the command would be as follows:

```
convert image1.jpg -resize 800x600 newimage1.jpg
```

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Passwordless SSH to remote machine

It can be really annoying (mostly in the enterprise environment) when you have to enter a password each time while doing an SSH to a remote machine. So, our aim here is to do a passwordless SSH from one machine (let's call it host A/User a) to another (host B/User b).

Now, on host A, if a pair of authentication keys are not generated for User a, then generate these with the following commands (do not enter a passphrase):

```
a@A:~> ssh-keygen -t rsa  
Generating public/private rsa key pair.  
Enter file in which to save the key (/home/a/.ssh/id_rsa):  
Created directory '/home/a/.ssh'.  
Enter passphrase (empty for no passphrase):  
Enter same passphrase again:  
Your identification has been saved in /home/a/.ssh/id_rsa.  
Your public key has been saved in /home/a/.ssh/id_rsa.pub.  
The key fingerprint is:
```

```
3e:4f:05:79:3a:9f:96:7c:3b:ad:e9:58:37:bc:37:e4 a@A
```

This will generate a public key on `/home/a/.ssh/id_rsa.pub`.

On host B as User b, create `~/.ssh` directory (if not already present) as follows:

```
a@A:~> ssh b@B mkdir -p .ssh  
b@B's password:
```

Finally, append User a's new public key to `b@B:~/.ssh/authorized_keys` and enter User b's password for one last time:

```
a@A:~> cat .ssh/id_rsa.pub | ssh b@B 'cat >> .ssh/authorized_keys'  
b@B's password:
```